MIL-S-003678A(DSA-DM)
26 April 1972
USED IN LIEU OF
MIL-S-3678
dated 15 January 1952

MILITARY SPECIFICATION

SHAKING MACHINE, LABORATORY (ROTATING TYPE)

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-S-3678. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers slide rotating apparatus for simulating hand shaking of micro slides, antigen flasks, etc., for flocculation tests in connection with cardiolipin syphilis tests.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Set-Up Paperboard.

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-M-7298	Manual, Technical: Commercial Equipment.

FSC 6640

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11 1969	Dimensions of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and suitable in all respects for the purpose intended.

3.1.1 Platform. Platform shall be of hard or half-hard cold rolled steel, covered with industrial sponge rubber or pinpoint molded rubber.

3.1.2 Springs. Springs shall be of corrosion-resisting steel or plated phosphor bronze.

3.1.3 Base. Base shall be of cold-or hot-rolled steel channel, coated with black asphalt varnish or grey vinyl lacquer.

3.1.4 Housing or guard. Housing or guard shall be of hard or half-hard rolled steel sheet with protective, baked-on enamel finish or baked hammertone enamel.

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3.2 Platform. Platform shall be a rubber-covered, floating platform, 13 inches ± 1 inch by 13 inches ± 1 inch, to accommodate fifteen 57 by 108 mm. Boerner slides. Metal shall be 1/8 inch thick and shall be securely, completely, and permanently covered on the top side with flat industrial sponge rubber, not less than 1/8 inch thick. Platform shall be attached to four corrosion-resisting, vertical springs, one near each corner of the bottom side, by means of corrosion-resisting nuts and bolts. The opposite end of each spring shall be secured near the corners of the metal base by similar nuts and bolts. Springs shall be made of 1/8 inch diameter wire, shall have approximately nine loops of 1-1/8 inches outer diameter, and shall be tapered or shaped at the ends to accommodate nuts and bolts securely. As an alternate, springs of uniform diameter throughout shall be used with press-fitted plated studs, firmly secured to the platform and base. Platform shall be supplied with four retaining bands of rubber for holding the slides to prevent creeping when rotator is in operation. Rubber covered top of platform shall present a coarse surface to minimize slippage of slides. As an alternate, the platform shall be made of 0.0478-inch thick (No. 18, United States Standard revised gage) metal, which shall be supported by a 1/8 inch ± 0.005 inch, cadmium-plated steel, "H" frame. The platform shall be covered with fabric-backed, 1/16-inch thick, non-slip, pinpoint rubber sheeting. Platform shall be attached to the "H" frame, which shall be supported by universal-jointed, zinc-plated, solid steel legs. Platform shall have four retaining walls plus an auxiliary bottle holder of stainless-steel wire.

3.3 Base. Base shall be a steel channel 12 inches wide by 2-1/4 inches by 1/4 inch thick and 12 inches long, nominal dimensions. As an alternate, the base shall consist of a 12-inch ± 0.032 inch square steel plate, 0.095 inch thick (No. 13, Birmingham of Stubs gage), with a 1/2-inch wide ± 1/16 inch steel rim to facilitate welding to the apron of the housing, and shall be finished with baked hammertone enamel.

3.4 Housing. Housing shall be a metal strip approximately 5 inches wide to engage the flanges of the base by means of suitable nuts and bolts. It shall enclose three sides and approximately 2/3 of back side so that sufficient area is exposed for accessibility to the interior of the apparatus. As an alternate, the housing shall be enclosed on all four sides, and access to the interior shall be accomplished by a readily removable platform. The front side bearing controls shall be secured to the base by means of a metal angle which shall be attached on the inside.

3.5 Motor. Motor shall be special, continuous duty, 115 volt, AC-DC, universal motor, with worm reduction gear. Worm and gear driving mechanism shall rotate the floating platform as specified in 3.11.1 to simulate hand rotation. Universal crank connection to the floating platform shall be made by means of a standard ball bearing, mechanically attached to the platform with shock-proof mounting, and shall require no oiling for maintenance, or the crank connection shall be attached to the "H" frame by means of a standard ball bearing.

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3.6 Controls. Controls shall be permanently mounted on the front side of the housing.

3.6.1 "Off-On" switch. "Off-On" switch shall be a suitable capacity toggle switch.

3.6.2 Speed control rheostat. Speed control rheostat shall consist, in part, of an easily adjustable pointer on a suitably graduated dial. The rheostat shall control the speed of the platform rotation as specified in 3.1.1.2.

3.6.3 Automatic cutoff timing device. Automatic cutoff timing device shall consist, in part, of an easily adjustable and securely mounted pointer on a dial which shall be graduated to indicate 0 to 28 minutes, in 1/2 minute subdivisions with at least every 5 minute interval numbered. Setting to a predetermined interval shall be accomplished by a simple adjustment of the pointer on the graduated dial. The timing device shall operate on both alternating and direct currents. The apparatus shall operate continuously and independently of the automatic cutoff timing device when the toggle switch is in the "on" position. The apparatus shall operate on the timing device when the toggle switch is in the "off" position.

3.7 Extension cord and plug. A three-conductor extension cord and plug of one-piece construction shall be supplied. A strain relief bushing shall be provided where the cord enters the housing. The cord shall be equipped with a three-blade male plug described in ANSI C73-11.

3.8 Rubber cushions. Four solid or suction rubber cushions shall be secured, one at each corner of the base, to act as legs.

3.9 Dimensions. Overall dimensions of apparatus shall be approximately 13 by 13 by 7-1/2 inches high.

3.9.1 Commercial tolerances shall apply to all gages specified herein.

3.10 Service Data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298 MANUAL, TECHNICAL, Commercial Equipment. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center

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(based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.11 Performance.

3.11.1 Platform rotation. Each point on platform shall rotate on a horizontal plane circling an area approximately 3/4 inch in diameter to simulate hand rotation.

3.11.2 Speed control. The apparatus shall be so regulated that when the pointer on the speed control rheostat is set at the lowest mark, the platform shall rotate 50 to 70 times per minute. Rotation speed of the platform shall increase progressively with the adjustment of the pointer to the higher marks until at the maximum mark, the platform shall rotate 210 $\pm$ 20 times per minute, when tested as specified in 4.4.1.

3.12 Workmanship. Workmanship shall be first class throughout. Rotator shall be free from defects which detract from its appearance or may impair its serviceability.

3.13 Identification marking. Each shaker shall have a plastic or nonferrous nameplate securely attached to one of the outer sides of the unit, bearing the following information:

Name or registered trademark of the manufacturer
Manufacturer's serial number
Electrical ratings (volts, amperes, watts and frequency)
Federal Stock Number
"U.S."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The shaker shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101	Component missing.
102	Finish or housing or guard not smooth or peeling.
103	Metal parts not free of rust or corrosion.
104	Rubber covering on platform not secure or completely affixed thereto.
105	Rubber covering on platform not provided with coarse surface to minimize slippage of slides.
106	Rubber cushions (feet) not secure or missing.
107	Shaking machine not free of sharp edges.

Minor

201	Control knobs not secure.
202	Manufacturer's identification marking not present, legible or complete.
203	Rubber covering on platform not free of tears or cuts.
204	Coating on base not completely applied or peeling.

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensionsal examination. Shaking apparatus shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4.1 Speed control test. Allow the shaking machine to run for ten minutes. Then count the number of rotations at the lowest speed setting point and at the highest speed setting point. Counting may be done by allowing the rotating table to touch the hand and, while watching the sweep second hand of a watch, record the number of oscillations every 15 seconds.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One shaking machine, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be packaged in a carton of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be packed in a container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table of PPP-B-636 shall apply for PPP-B-636 box.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be packed in a container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table of PPP-B-636 shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Federal Stock Number (FSN).

6.2 This specification is applicable to the following item listed in the Federal Supply Catalog:

Federal Stock Number	Item Identification
6640-438-9600	SHAKING MACHINE, LABORATORY, Slide, 115 volt, AC-DC

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

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Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
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Review information is current as the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

MIL-S-003678A(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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